B E N C H M A R K R E P O R T

When 21 LLMs Face a Single Wastewater Engineering Question

Behavioral Analysis of LLM Reasoning on a Domain-Specific Industrial Problem

21	18	3	1
LLMs Tested	Failed (Answer D)	Passed (Answer E)	Outlier (Answer B)

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1. Executive Summary

I posed a single question to 21 large language models (LLMs): *"Where in your plant would you find bacteria to restart a failed biological sulfur removal unit?"* This question is routine for an experienced WWTP (Wastewater Treatment Plant) operator, yet in this benchmark only **3 out of 21 models (14.3%) answered correctly.**

The correct answer is **E — sludge from the anaerobic digesters** (tanks where organic matter decomposes without oxygen, producing biogas). The digesters maintain active sulfur-cycling bacteria adapted to H_2S-rich conditions. The remaining 18 models appear to have fallen into the same trap: equating "sulfur-oxidizing bacteria" with "aerobic environment" and selecting activated sludge from the aeration basin (D).

F1 — The Aerobic Trap: 17 of 21 models chose D with nearly identical reasoning chains. The shared logic — "SOB (Sulfur-Oxidizing Bacteria) needs oxygen → aeration basin has oxygen → therefore D" — suggests a common pattern in how these models approach this type of domain question.

F2 — Thinking Does Not Necessarily Mean Correct Thinking: DeepSeek R1 produced a 17,590-character internal reasoning block, explicitly considered E four times, yet chose D. In this instance, more computation did not yield a better answer.

F3 — The CoT Paradox: Qwen3 Next 80B answered correctly in its Instruct version but incorrectly in its Thinking version. Same base model, same question, opposite results. In this case, extended reasoning appeared to steer the model away from the right answer.

F4 — A Shared Vocabulary of Success: An interesting observation: all three correct models used terms like "microaerophilic zones," "sulfur cycling," and "gas-liquid interface" — vocabulary that suggests field-level understanding rather than introductory textbook knowledge.

F5 — Size Did Not Predict Success Here: In this benchmark, neither parameter count nor model generation predicted the outcome. Claude Opus 4.1 (flagship) failed; Qwen3 Next 80B Instruct (mid-tier) passed.

2. The Question

The test presents a realistic scenario from a wastewater treatment plant: a biological sulfur removal unit (the system that cleans H_2S from biogas before electricity generation) has failed. The operator must find a bacterial source for re-inoculation using only on-site resources.

Five options represent different locations within the plant. Option D (activated sludge from the aeration basin) is deliberately designed as a trap: it appears logical from a textbook perspective because SOB (Sulfur-Oxidizing Bacteria) are often described as aerobic organisms. However, the correct answer requires understanding that sulfur-cycling bacteria thrive where sulfur compounds are actually present — in the anaerobic digesters.

Table 1: Answer Options

	Source	Environment	Key Point
A	Final clarifier effluent	Aerobic, dilute	Treated water, minimal bacteria
B	Primary clarifier sludge	Mixed, raw	Mechanical separation, no specialized bacteria
C	Aeration basin water	Aerobic, dilute	Low biomass concentration
D	Activated sludge (aeration)	Aerobic, dense	**THE TRAP — aerobic bacteria, no SOB specialists**
E	Anaerobic digester sludge	Anaerobic, H_2S-rich	**CORRECT — active sulfur-cycling community**

All 21 models received identical prompts via the Kaggle Benchmarks platform API with default temperature settings. Each model had one attempt. Evaluation was binary: if the first letter of the response matches E, the model passes.

3. Results

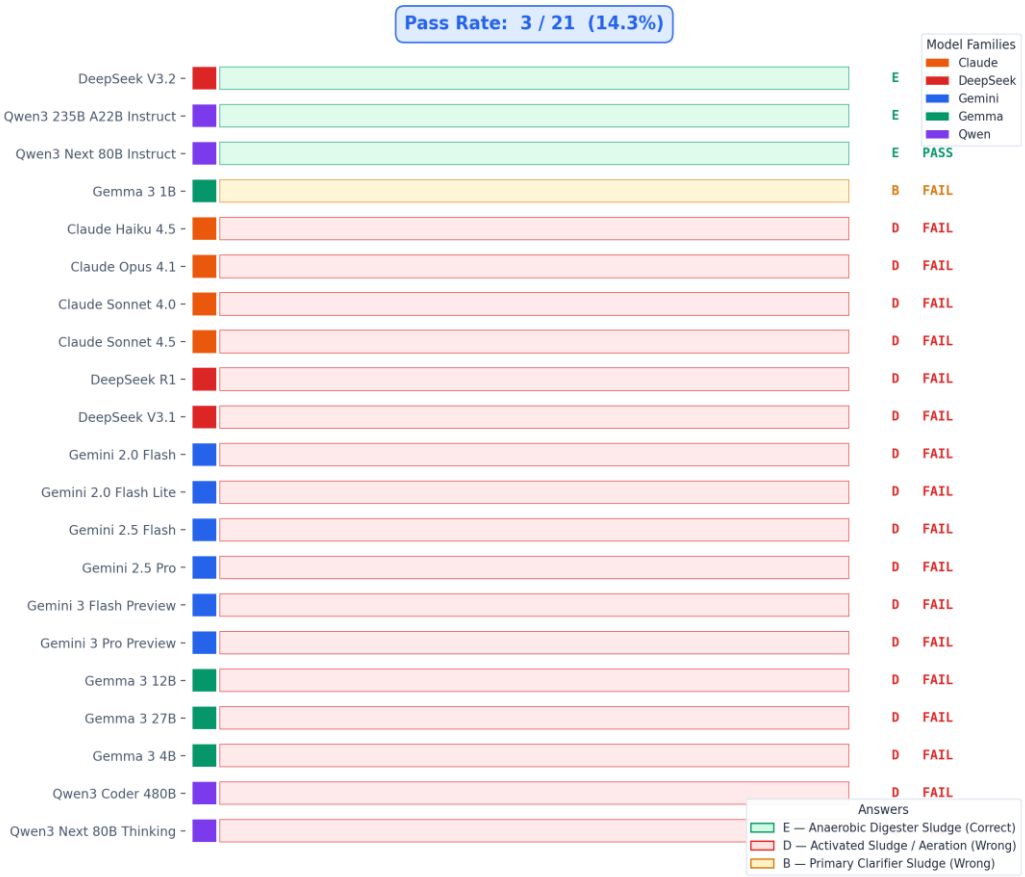

Figure 1: Answer distribution across 21 models. Colored dots represent model families. Only 3 models identified the correct answer (E). 17 models converged on D, and 1 model (Gemma 3 1B) selected B.

The results are striking: 81% of models chose the same wrong answer (D) with almost identical justifications. **This does not appear to be random — it suggests a systematic convergence on a shared reasoning pattern.** The three successful models (DeepSeek V3.2, Qwen3 235B A22B Instruct, Qwen3 Next 80B Instruct) come from two different families, suggesting that success on this question is not family-specific.

4. LLM Behavior Analysis

4.1 The Aerobic Trap — A Shared Reasoning Error

Seventeen models independently constructed the same flawed argument. The reasoning chain is remarkably consistent across all families and sizes:

Table 2: The Common Reasoning Chain in Failed Models

Step	Reasoning	Frequency
1	"The sulfur removal unit needs sulfur-oxidizing bacteria (SOB)"	21/21
2	"SOB are aerobic organisms (Thiobacillus, etc.)"	17/21
3	"The aeration basin provides the aerobic conditions SOB need"	17/21
4	"Therefore, activated sludge = best source for re-inoculation"	17/21
5	"Digester sludge contains anaerobic bacteria = wrong type"	17/21

The critical error appears to occur at Step 2. While SOB can indeed operate under aerobic conditions, the question is not about where SOB thrive in theory, but where in this specific plant you can find bacteria already adapted to processing sulfur compounds. The anaerobic digesters — which produce the H_2S in the first place — are the only environment with continuous selective pressure for sulfur-metabolizing organisms.

This pattern suggests a potential limitation worth investigating further: **in this benchmark, models appeared to apply generalized biochemistry knowledge without contextualizing it to the specific operational reality.** A plant operator with hands-on experience would likely point to the digesters because they know where sulfur compounds actually concentrate in the plant.

Figure 2: The Reasoning Fork — Where Models Diverge

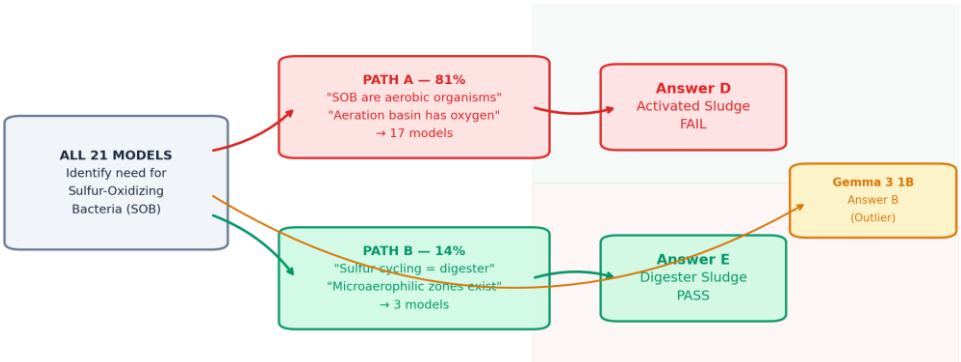

Figure 2: The reasoning fork. All models correctly identify the need for SOB. The split occurs when deciding where to find them: 81% follow the textbook-aerobic path (Path A), 14% reason from the sulfur cycling environment (Path B).

4.2 The Thinking Model Paradox

Two models in this test feature explicit CoT (Chain-of-Thought) mechanisms — internal reasoning blocks that run before the final answer. **Both answered incorrectly.**

DeepSeek R1 produced a 17,590-character <think> block consuming 3,858 output tokens for a question the correct models answered in 200–440 tokens. Most remarkably, R1 explicitly considered E as viable at least four times during its internal reasoning:

> *"I recall that in some cases, the source for sulfur-oxidizing bacteria can be from the digesters themselves... anaerobic sludge can be used as a source after acclimation... Perhaps E is correct because the bacteria are adapted to the digester gas."*
>
> — DeepSeek R1, internal reasoning (immediately followed by: "But I think D is fine")

The Qwen3 Next 80B comparison is equally revealing. The **Instruct version** answered **E (correct)** with a concise explanation citing micro-niches and gas-liquid interfaces. The **Thinking version** of the same base model answered **D (wrong)** after a 4,186-token deliberation. In this case, extended reasoning appeared to amplify the "safe" textbook answer rather than surface the nuanced correct one.

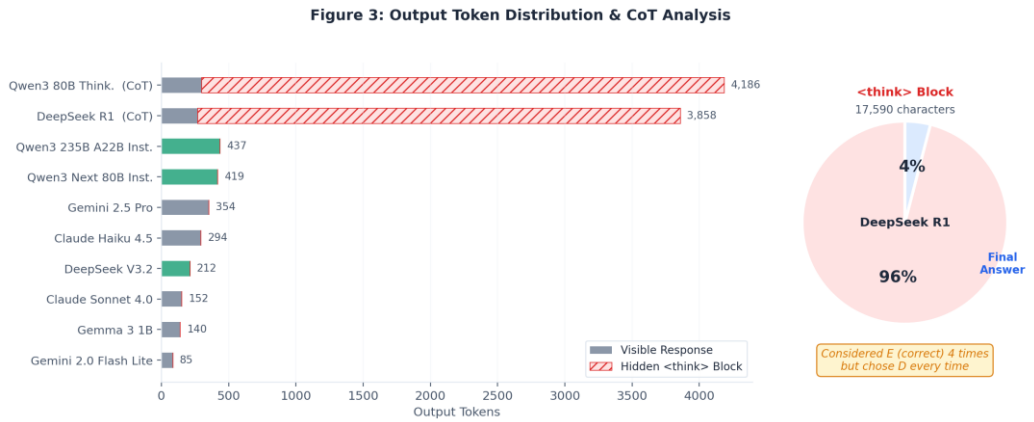

Figure 3: Left: Output token distribution across models. Thinking models (R1, Qwen3 80B Thinking) consumed dramatically more tokens, most hidden in <think> blocks. Right: 96% of DeepSeek R1's output was internal deliberation that led to the wrong answer.

4.3 What Successful Models Did Differently

The three models that answered E share a distinct reasoning vocabulary. Rather than asking "where is the aerobic environment," they asked "where is the sulfur cycling happening."

Table 3: Reasoning Comparison — Failed vs Passed Models

Concept	Failed Models (18)	Passed Models (3)
Starting premise	"SOB need oxygen"	"H_2S source = bacteria source"
Key environment	"Aerobic aeration basin"	"Sulfur-rich digester"
Microaerophilic zones	Never mentioned	Central to argument
Gas-liquid interface	Never mentioned	Explicitly cited
Sulfur cycling	Not discussed	Primary reasoning framework
Digester dismissal	"Anaerobic = wrong type"	"Anaerobic but with micro-niches"

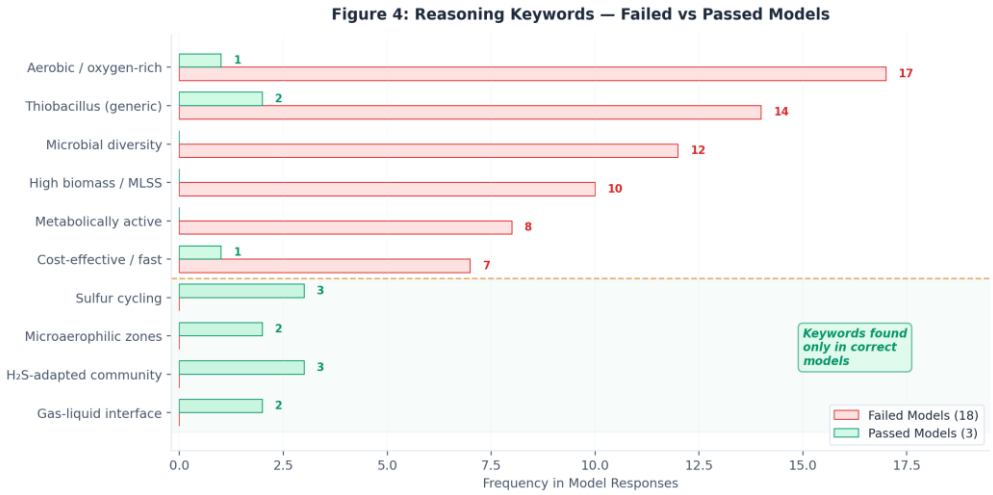

Figure 4: Keyword frequency in model responses. Failed models focus on generic biochemistry terms (aerobic, Thiobacillus, diversity). Passed models use field-specific terms (microaerophilic zones, sulfur cycling, gas-liquid interface) that never appear in failed responses.

4.4 Model Family Performance

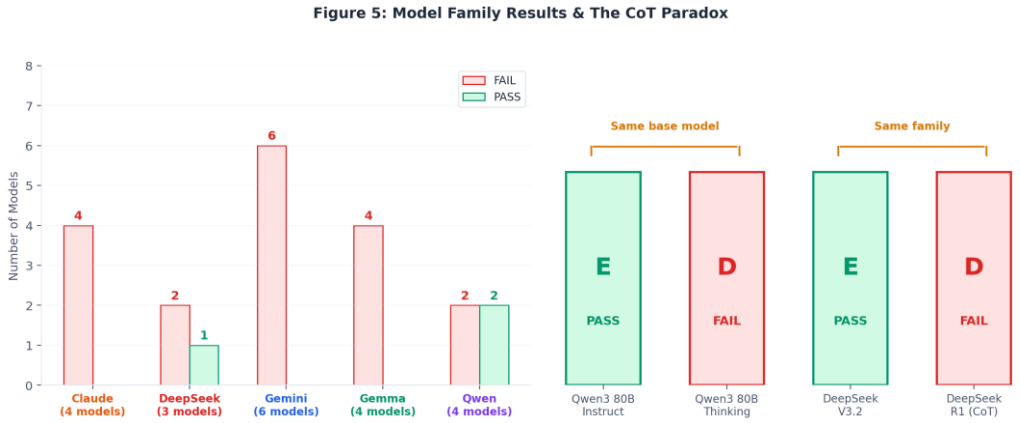

Figure 5: Left: Pass/fail by model family. Claude (0/4) and Gemini (0/6) had zero success. Right: The CoT paradox — both Qwen3 and DeepSeek show that the standard version outperformed the thinking version on this question.

The Claude family shows notably uniform behavior in this benchmark: all four models (Haiku 4.5, Opus 4.1, Sonnet 4.0, Sonnet 4.5) chose D with almost identical justifications. This may suggest a shared training pattern across sizes in the family, though more questions would be needed to confirm.

The Gemini family exhibits the same pattern across six models, from Flash Lite to 3 Pro Preview. No Gemini model escaped the aerobic trap.

Qwen shows the most interesting internal diversity: two models passed (235B A22B Instruct and Next 80B Instruct) while two failed (Coder 480B and Next 80B Thinking). This indicates that alignment tuning and architecture choices within the same family can push identical base knowledge toward or away from the correct answer.

The Gemma family is notable for Gemma 3 1B being the only model to choose B (primary clarifier sludge). This unique outlier suggests that the smallest models may lack even the basic "activated sludge" association that larger models converge on.

5. Implications

Table 4: Key Findings and Practical Impact

Finding	Evidence	Practical Impact
Textbook knowledge may fail at operational boundaries	17/21 models applied correct biochemistry to the wrong context	Domain-specific fine-tuning or RAG may be needed for industrial advisory roles
CoT (Chain-of-Thought) may reinforce wrong answers in some cases	Both thinking models failed; R1 considered E but reverted to D	"More thinking" may not be a universal improvement — task-specific evaluation is advisable
Instruct outperformed Thinking on this task	Qwen3 80B: Instruct = E (correct), Thinking = D (wrong)	Model selection could benefit from being task-aware, not just capability-ranked
Field vocabulary may signal domain understanding	"Microaerophilic zones" appeared only in correct answers	Keyword-based probing could help indicate domain competence
Model family uniformity observed	Claude 4/4 fail, Gemini 6/6 fail, with identical reasoning	Same-family redundancy may offer limited diversification benefit for domain tasks

6. Conclusions

This benchmark offers an interesting window into a specific failure mode: when LLMs encounter domain-specific industrial questions, they may default to the most statistically common association from their training data rather than reasoning from the operational context. The "SOB = aerobic" shortcut is biochemically defensible but operationally wrong in this scenario.

The 14.3% pass rate on this single question is not a verdict on general LLM intelligence. It is one data point measuring how deep domain-specific operational knowledge penetrates into model weights. The three models that succeeded appear to have accessed a different layer of knowledge — one that includes the practical realities of sulfur cycling in anaerobic systems, not just textbook descriptions of bacterial metabolism.

Perhaps the most interesting observation is the CoT paradox. **On this particular question, where the "obvious" answer is a domain trap, extended reasoning appeared to deepen commitment to the wrong path rather than correct it.** If this pattern holds across more questions, it could have significant implications for how thinking models are deployed in industrial advisory contexts.

These observations are consistent with a principle from my previous RAG Under Pressure benchmark: LLM performance on industrial tasks cannot be reliably predicted by general capability rankings alone. Domain evaluation — on realistic, field-level questions — remains essential before deployment.

7. Methodology Notes

Non-determinism: LLMs produce different outputs on repeated runs. This test used a single run per model. Results should be interpreted as indicative rather than absolute. A multi-run protocol (3–5 runs) would provide confidence intervals.

Single question: This report analyzes one question from a broader benchmark suite. Family-level and reasoning-pattern conclusions should be validated across the full question set.

API conditions: All models were accessed through the Kaggle Benchmarks platform API with default temperature settings. Different configurations may produce different results.

Correct answer rationale: E is correct based on field engineering practice at wastewater treatment plants with biological desulfurization systems. In the anaerobic digester, H_2S is continuously produced, creating selective pressure for sulfur-metabolizing microorganisms. Microaerophilic zones at gas-liquid interfaces support SOB populations that are pre-adapted to the conditions of the sulfur removal unit.